

第 7 章 Laplace 变换

§ 7.1 基本概念

① 定义

设 $f(t)$ 是 $[0, +\infty)$ 的实(复)值函数, 若对参数 $s = c + iy$

$F(s) = \int_0^{+\infty} f(t) e^{-st} dt$ 在某一区域内收敛, 则称其为 $f(t)$ 的 Laplace 变换

$$\mathcal{L}[f(t)] = \underbrace{F(s)}_{\text{象函数}} = \int_0^{+\infty} f(t) e^{-st} dt$$

而 $f(t)$ 称为 $F(s)$ 的 Laplace 逆变换, 记为 $f(t) = \mathcal{L}^{-1}[F(s)]$

常用 1

$$\text{单位阶跃函数 } u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases} \quad \mathcal{L}[u(t)] = \frac{1}{s} \quad (\operatorname{Re}(s) > 0 \text{ 时收敛})$$

常用 2

$$\mathcal{L}[e^{kt}] = \frac{1}{s-k} \quad \operatorname{Re}(s) > \operatorname{Re}(k)$$

② 存在定理

若 $f(t)$ 满足 $\begin{cases} \text{在 } t \geq 0 \text{ 的任一有限区间上分段连续} \\ t \rightarrow +\infty, \text{ 增长速度不超过指数函数 } |f(t)| \leq M e^{ct} \end{cases}$

则 Laplace 变换在 $\operatorname{Re}(s) > c$ 上存在且解析

常用 3

$$f(t) = t^a \quad (a > -1) \quad \mathcal{L}[f(t)] = \frac{\Gamma(a+1)}{s^{a+1}} \quad \begin{matrix} \operatorname{Re}(s) > 0 \\ a \text{ 为非负整数 } n \text{ 时 } \mathcal{L}[f(t)] = \frac{n!}{s^{n+1}} \end{matrix}$$

§ 7.2 基本性质

① 线性性质

$$\mathcal{L}[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(s) + a_2 F_2(s)$$

$$\mathcal{L}^{-1}[a_1 F_1(s) + a_2 F_2(s)] = a_1 f_1(t) + a_2 f_2(t)$$

常用 4 $\mathcal{L}[\sin \omega t] = \frac{\omega}{s^2 + \omega^2} \quad \mathcal{L}[\cos \omega t] = \frac{s}{s^2 + \omega^2} \quad \operatorname{Re}(s) > 0$

[例 1] $\mathcal{L}^{-1}\left[\frac{s}{(s+2)(s+4)}\right]$

$$\because \frac{s}{(s+2)(s+4)} = 2 \frac{1}{s+4} - \frac{1}{s+2}$$

$$\therefore \mathcal{L}^{-1}\left[\frac{s}{(s+2)(s+4)}\right] = 2e^{-4t} - e^{-2t}$$

② 平移性质

时移性 $\mathcal{L}[f(t-t_0)] = e^{-st_0} F(s)$

频移性 $\mathcal{L}[e^{s_0 t} f(t)] = F(s-s_0)$

[例 2] 求 $\mathcal{L}(e^{at} \sin \omega t)$

$$\because \mathcal{L}(\sin \omega t) = \frac{\omega}{s^2 + \omega^2}$$

$$\therefore \text{由频移性, } \mathcal{L}(e^{at} \sin \omega t) = \frac{\omega}{(s-a)^2 + \omega^2}$$

③ 微分性质

原象函数 $\mathcal{L}[f'(t)] = sF(s) - f(0^+)$

$$\mathcal{L}[f^{(n)}(t)] = s^n F(s) - s^{n-1} f(0^+) - s^{n-2} f'(0^+) - \cdots - f^{(n-1)}(0^+)$$

象函数 $\mathcal{L}^{-1}[F(s)] = (-t)^n f(t) \longleftrightarrow \mathcal{L}[t^n f(t)] = (-1)^n F^{(n)}(s)$

[例 3] $\mathcal{L}[t^2 \cos kt]$

$$\because \mathcal{L}[\cos kt] = \frac{s}{s^2 + k^2}$$

$$\therefore \mathcal{L}[t^2 \cos kt] = (-1)^2 \left(\frac{s}{s^2 + k^2} \right)'' = \frac{2s^3 - 6k^2 s}{(s^2 + k^2)^3}$$

④ 积分性质

$$\text{原象函数 } \mathcal{L} \left[\int_0^t f(u) du \right] = \frac{F(s)}{s}$$

$$\text{象函数 } \mathcal{L} \left[\frac{f(t)}{t} \right] = \int_s^{+\infty} F(u) du$$

$$\text{[例 4]} \quad \mathcal{L} \left[\int_0^t \cos t dt \right] = \frac{1}{s} \mathcal{L} [\cos t] = \frac{1}{s} \frac{s}{s^2+1} = \frac{1}{s^2+1}$$

$$\text{[例 5]} \quad \mathcal{L} \left[\frac{\sinh t}{t} \right] = \int_s^{+\infty} \frac{1}{u^2-1} du = \frac{1}{2} \ln \frac{u-1}{u+1} \Big|_s^{\infty} = \frac{1}{2} \ln \frac{s+1}{s-1}$$

⑤ 极限性质

有条件!

$$f(0^+) = \lim_{s \rightarrow \infty} s F(s) \quad f(\infty) = \lim_{s \rightarrow 0} s F(s) \quad \swarrow$$

$$\text{[例 6]} \quad \text{如果 } \mathcal{L}[f(t)] = \frac{1}{s+a} \quad (a>0), \text{ 求 } f(0), f(\infty)$$

$$f(0) = \lim_{s \rightarrow \infty} \frac{s}{s+a} = 1 \quad f(\infty) = \lim_{s \rightarrow 0} \frac{s}{s+a} = 0$$

⑥ 卷积性质

定义: 若 $\int_{-\infty}^{+\infty} f_1(\tau) f_2(t-\tau) d\tau$ 存在, 则称为 $f_1(\tau)$ 与 $f_2(\tau)$ 的卷积 $f_1(\tau) * f_2(\tau)$

满足 交换律、结合律、分配律

$$\text{若 } f_1(t) = f_2(t) \equiv 0 \quad (t < 0), \text{ 则 } f_1(t) * f_2(t) = \int_0^t f_1(\tau) f_2(t-\tau) d\tau$$

$$\text{[例 7]} \quad \text{已知 } t < 0 \text{ 时 } f_1(t) = f_2(t) \equiv 0, \text{ 求 } f_1(t) * f_2(t):$$

$$(1) \quad t * \sin t = \int_0^t \tau \sin(t-\tau) d\tau = t - \sin t$$

$$(2) \quad t * e^{at} = \int_0^t \tau e^{t-\tau} d\tau =$$

$$(3) \quad \sin \omega t * \sin \omega t = \int_0^t \sin \omega \tau \sin[\omega(t-\tau)] d\tau =$$

$$\text{如果 } \mathcal{L}[f_1(t)] = F_1(s), \mathcal{L}[f_2(t)] = F_2(s)$$

$$\text{则 } \mathcal{L}[f_1(t) * f_2(t)] = F_1(s) F_2(s)$$

$$\mathcal{L}^{-1}[F_1(s) F_2(s)] = f_1(t) * f_2(t)$$

[例 8] 若 $\mathcal{L}[f(t)] = \frac{1}{(s^2+4s+13)^2}$, 求 $f(t)$

$$\begin{aligned} \text{解 } \because F(s) &= \frac{1}{9} \times \frac{3}{(s+2)^2+3^2} \times \frac{3}{(s+2)^2+3^2} \\ &\quad \frac{3}{(s+2)^2+3^2} \xrightarrow{\mathcal{L}^{-1}} e^{-2t} \sin 3t \\ \therefore f(t) &= \frac{1}{9} (e^{-2t} \sin 3t) * (e^{-2t} \sin 3t) \\ &= \dots \end{aligned}$$

§ 7.3 Laplace 逆变换

一般式 $f(t) = \frac{1}{2\pi j} \int_{\beta-j\omega}^{\beta+j\omega} F(s) e^{st} ds$ ($\int_0^{+\infty} e^{-\beta t} f(t) u(t) dt$ 收敛)

定理: 若 $F(s)$ 在全平面只有有限个奇点 $s_1, \dots, s_n$ (均在 $\operatorname{Re} s = \beta$ 左侧)

且 $\lim_{s \rightarrow \infty} F(s) = 0$, 则 $t > 0$ 时

$$f(t) = \sum_{k=1}^n [F(s) e^{st}; s_k]$$

[例 1] $F(s) = \frac{1}{s(s-1)^2}$

[例 2] $F(s) = \frac{1}{s^2(s+1)}$

$$F(s) = \frac{1}{s^2(s^2+1)}$$

$$F(s) = \frac{1}{(s^2+2s+5)^2}$$

§ 5 应用: 解 ODE

[例 1] 解 $\begin{cases} x''(t) - 2x'(t) + 2x(t) = 2e^t \cos t \\ x(0) = x'(0) = 0 \end{cases}$

令 $X(s) = \mathcal{L}[x(t)]$:

$$s^2 X(s) - s x(0) - x'(0) - 2(s X(s) - x(0)) + 2 X(s) = \frac{2(s-1)}{(s-1)^2+1}$$

$$X(s) = \frac{2(s-1)}{[(s-1)^2+1]^2}$$

$$x(t) = \mathcal{L}^{-1} \left[\frac{2(s-1)}{[(s-1)^2+1]^2} \right] = e^t \mathcal{L}^{-1} \left[\frac{2s}{s^2+1} \right] = -e^t \mathcal{L}^{-1} \left[\left(\frac{1}{s^2+1} \right)' \right] = -(-t)' e^t \mathcal{L}^{-1} \left(\frac{1}{s^2+1} \right) = te^t \sin t$$